

Viswa Subramanian

+1(858) 405 8352 viswasubramanian055@gmail.com [linkedin.com/in/viswasubramanian555](https://www.linkedin.com/in/viswasubramanian555) [viswa-005.github.io](https://github.com/viswa-005)

SUMMARY: Embodied AI & Robotics Engineer focused on robot learning, Vision-Language-Action models, and real-world manipulation. Experience spanning VLA fine-tuning, interpretability & steering, learned perception, closed-loop control, and simulation-to-real robotics, with hands-on deployment across robot arms and embodied AI systems.

EXPERIENCE

TACTUS AI, San Diego | Autonomous Lab Robots | Embodied AI Intern (Ongoing) Jul - Oct 2026

- Developed **6-DoF pose** visual servoing pipeline with closed-loop arm control achieving **5 mm** end-effector position accuracy
- Built a robot-agnostic SE(3) trajectory-correction system to compensate for mobile-base docking errors; validated successful grasps after base displacements up to **30 mm** translation and **5°** yaw, with residual positioning errors to **1–5 mm**
- Achieved **0.5 mm** pose repeatability in robot trials using inverse kinematics, joint-sag compensation, diagnosed camera bias
- Evaluated MuJoCo workflows for controller errors, sim-to-real discrepancies manipulation reachability & compensation

Advanced Robotics & Controls Lab | Graduate Researcher | Dr. Jaeyoung Huh, UCSD Feb - Jul 2026

- Built **$\pi 0.5$ Droid** interpretability and causal-steering pipelines on **NVIDIA A100 GPU** using JAX, including hidden-state probing, success/failure representation directions, and producing measurable action shifts
- Fine-tuned $\pi 0.5$'s action expert with **LoRA** on **NYU Franka** demonstrations, reducing open-loop action **MAE ~ 82%**, and used **ActionLens** probes to study where robot actions become geometrically decodable across hidden representations

Coordinated Robotics Lab | Graduate Researcher | Prof. Thomas R Bewley, UCSD Jan - Mar 2026

- Developed a multi-task MobileNetV2 model achieving **98%** classification accuracy and **$\leq 2.8^\circ$ MAE** for joint-angle estimation
- Designed a circular regression representation for angular prediction, eliminating discontinuities at the $0^\circ/360^\circ$ boundary
- Deployed the model as **ONNX FP32** on Raspberry Pi 5 achieving **~45ms** per inference in a gas pipe robot for **GTI Energy**

PROJECTS

State2Pose: Start-Pose Selection for a Frozen VLA from Its Own Hidden State

$\pi 0.5$ | LIBERO | MuJoCo | JAX | OpenPI | <https://github.com/viswa-5/State2Pose>

- Investigated $\pi 0.5$ LIBERO-10 sensitivity by building a **3,400-rollout, 40-scenes** with per-rollout hidden-state logging, identifying wrist-start-pose variations as driving performance between **0% - 57% success**
- Trained KV-state probes on $\pi 0.5$'s internal representations **to rank candidate wrist poses**, outperforming external image features (0.67 vs. 0.52 AUROC) and improving held-out scene success from **59% to 78% without fine-tuning**
- Built a closed loop that visits candidate wrist poses and re-scores each from $\pi 0.5$'s own hidden state reaches **69 % success on 30 unseen LIBERO-10 scenes**, against **65%** for the best pose & **51%** for never moving, with **no change to policy weights**

Flow-Matching SDF Completion for Robot Perception Diffusion | SDF

- Re-engineered a Diffusion-SDF 3D shape-completion pipeline by replacing iterative **DDPM** sampling with **Flow Matching**, achieving **19.4x** faster 3D shape generation while preserving completion quality in a **7x** reduced compute

Failure-Recovery Dataset for VLAs $\pi 0.5$ | RoboCasa | Dataset

- Building the **long-horizon failure recovery dataset** for VLAs in simulation: injecting artificial and inference failures (missed grasp, mid-transport drop) and scripted retry-until-success, with per-frame phase labels (nominal / error / recovery)

EDUCATION

UNIVERSITY OF CALIFORNIA SAN DIEGO, CA, USA

Sep 2024 - Jun 2026

Master of Science, Mechanical and Aerospace Engineering (Robotics & AI)

PSG COLLEGE OF TECHNOLOGY, TAMIL NADU, INDIA

Oct 2020 - May 2024

Bachelor of Engineering, Robotics and Automation Engineering

SKILLS

AI Focus : Fine-tuning, Model Optimization, Multimodal AI, Edge AI, Diffusion Models, VLM, VLA, WAM, V-JEPA

Robotics Focus : Pose Estimation, Sensor Fusion, Depth Estimation, 3D Reconstruction, SLAM, Visual Servoing

Languages & Tools: Python, C++ , MATLAB, Git, Linux, Docker, Kubernetes, ROS, Gazebo, MuJoCo

Libraries & Frameworks: PyTorch, TensorFlow, JAX, NumPy, Pandas, Matplotlib, OpenCV, ONNX, TensorRT

Hardware : LiDAR, Intel Realsense, OakD; RTK GNSS; Arduino, ESP32, Raspberry Pi; UR5, OpenArm, FANUC arms